

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i>		
Student Name:	A.Veerasanthosh	Reg. No.:	192525187
Section / Batch:		Date:	10/07/2026

Code of Conduct

I **A.Veerasanthosh (192525187)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

A.Veerasanthosh

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure / Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 - 6	20

Section / Topic	Focus	Q Nos	Marks
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

Q1 – C Code – Array Traversal

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QUESTIONS

Q1
C Code – Array Traversal
10 marks**Q2**
C Code – Linear Search
10 marks**Q3**
C Code – Binary Search Mid
10 marks**Q4**
C Code – Pointer with Array
10 marks**Q5**
C Code – Array Update
10 marks**Q1 C Code – Array Traversal** 10 marks

```
#include <stdio.h>

int main() {
    int arr[] = {5, 10, 15, 20};
    int i, sum = 0;
    for(i = 0; i < 4; i++)
        sum += arr[i];
    printf("%d", sum);
    return 0;
}
```

Your Answer

50

Q2 – C Code – Linear Search

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Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

```
#include <stdio.h>

int main() {
    int arr[] = {2,4,6,8,10};
    int key = 8;
    int i;
    for(i=0;i<5;i++){
        if(arr[i]==key){
            printf("%d",i);
            break;
        }
    }
}
```

Your Answer

3

Q3 – C Code – Binary Search Mid

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Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Your Answer
3

Q4 – C Code – Pointer with Array

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Q1 ✓
C Code – Array Traversal
10 marks

Q2 ✓
C Code – Linear Search
10 marks

Q3 ✓
C Code – Binary Search Mid
10 marks

Q4 ✓
C Code – Pointer with Array
10 marks

```
#include<stdio.h>

int main(){
    int arr[]={10,20,30};
    int *p=arr;
    printf("%d",*(p+2));
}
```

Your Answer

30

Q5 – C Code – Array Update

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Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update

```
#include<stdio.h>

int main(){
    int arr[]={1,2,3};
    arr[1]=arr[0]+arr[2];
    printf("%d",arr[1]);
}
```

Your Answer
4

Q6 – C Code – Linked List Node

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Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code – Linked List Node
10 marks

```
#include<stdio.h>

struct Node{
    int data;
    struct Node *next;
};

int main(){
    struct Node n1,n2;
    n1.data=10;
    n2.data=20;
    n1.next=&n2;
    n2.next=NULL;
    printf("%d",n1.next->data);
}
```

Your Answer
20

Q7 – C Code – Time Complexity Loop

Q7 C Code - Time Complexity Loop

10 marks

```
for(i=0;i<n;i++)  
printf("%*");  
The complexity is
```

Your Answer

$O(n)$

Q8 – C Code – Nested Loop

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Q8 C Code - Nested Loop

10 ma

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("*");
```

Your Answer

$O(n^2)$

Q9 – C Code – Binary Search Complexity

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Q9 C Code - Binary Search Complexity

10 marks

```
while(low <= high){  
    mid = (low + high) / 2;  
}
```

Your Answer

$O(\log n)$

Q10 – C Code – Pointer Increment

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Q10 C Code - Pointer Increment

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10